ALICE L WALTON SCHOOL OF MEDICINE

Ventricular Arrhythmias

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Adapted from: Ventricular Arrhythmias and Junctional Escape Rhythms

Listen to this Brick

0:00 / 23:43 1x

Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Describe the pathophysiology of ventricular arrhythmias
2. Describe the clinical findings associated with ventricular arrhythmias
3. Recognize the characteristic features of QTc prolongation
4. Outline a tre[illegible]s

CASE CONNECTION

You are asked to evaluate GJ, a 74-year-old male who had a coronary artery bypass graft (CABG) procedure 1 day ago. When you enter his room, GJ looks very worried and says, "I'm having trouble breathing. It feels like I'm suffocating." His heart rate is 150 beats/min and regular, respiratory rate is 20 breaths/min, and blood pressure is 90/60 mm Hg. As you review the telemetry tracings, which show a wide complex tachycardia, GJ loses consciousness. You immediately call a code blue and begin the advanced cardiovascular life support (ACLS) protocol.

What will be the next step in GJ's treatment? How will you explain to him what happened? Consider your answers as you read, and we'll revisit GJ at the end of the brick.

GO TO CONCLUSION ↓

What Are Ventricular Arrhythmias? (LO#1–3)

Ventricular arrhythmias originate at the level of the ventricle, as opposed to supraventricular arrhythmias, which arise at or above the AV junction. Here we'll cover ventricular arrhythmias, including the mostly harmless premature ventricular contractions (PVCs) as well as other potentially lethal ones (VT and VF), followed by accelerated idioventricular rhythm.

What Are Premature Ventricular Contractions?

Starting with the most common and most harmless ventricular arrhythmias, PVCs are single abnormal beats experienced by patients as an extra heartbeat or "skipping a beat." They are just as the name indicates: premature (occurring earlier in the cardiac cycle than expected) and ventricular (having their origin in one of the ventricles).

Clinical Presentation

PVCs can be asymptomatic, or patients may complain of a skipped beat. The sensation of a "skip" is due to pause following the premature beat, before normal sinus rhythm resumes. Less commonly, patients may have light headedness or dizziness.

Pathophysiology

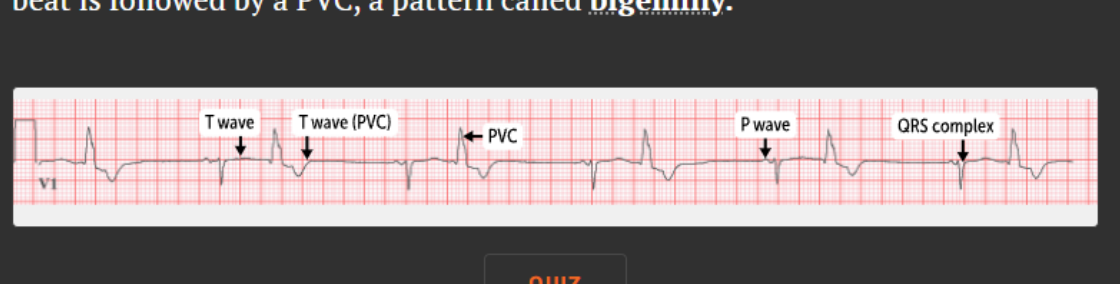
While PVC's may be a sign of underlying structural heart disease such as MI, they are often seen in otherwise normal individuals with no evidence of structural heart disease. The mechanisms include re-entry and enhanced automaticity. (Review the Pathophysiology brick).

Secondary Causes. PVCs are often worsened by alcohol, stimulant drugs such as cocaine and amphetamines, caffeine, or tobacco. Some patients note increased frequency during exercise or when they are anxious.

Diagnosis

PVCs are diagnosed on ECG: they typically show up as a single abnormal beat (premature, occurring earlier in the cardiac cycle than normal) with a widened QRS complex, a duration of more than 120 msec and an abnormal QRS morphology. A compe[illegible]he PVC (no more electrical impulses for a period twice the regular P-P interval). Furthermore, ST segments and T waves in PVCs usually point the opposite direction from the major part of the QRS complex.

PVCs sometimes recur with regularity. For example, in Figure 1, each normal beat is followed by a PVC, a pattern called **bigeminy**.



QUIZ

Figure 1 Premature ventricular contractions in pattern of bigeminy

Treatment

Because PVCs are not in and of themselves life threatening, treatment is largely directed toward avoidance of alcohol, tobacco, caffeine, or other precipitating drugs. β-Blockers can reduce the frequency in patients who are symptomatic and do not respond to lifestyle changes.

What pattern is described as a bigeminy in an ECG?

What Are Ventricular Tachycardia and Fibrillation?

Ventricular tachycardia (VT) and ventricular fibrillation (VF) are dangerous arrhythmias that occur from diseased myocytes within the ventricles. They are most commonly caused by MI or other underlying heart disorders, such as cardiomyopathy.

Clinical Presentation

In general, VT has a better prognosis than VF and a range of presentations from moderate symptoms to full cardiac arrest. In VT, the heart still makes coordinated contractions at a rate of 100 to 250 beats/min. However, the high rate does not allow enough time for the heart to relax enough to fill with blood between contractions. The cardiac output then drops, leading to syncope or dizziness. Blood can back up into the lungs, leading to pulmonary edema with shortness of breath. They might also feel chest pain or the rapid beat of the heart as palpitations. If the rate is too high, the blood pressure may drop, and the patient becomes unconscious and has a cardiac arrest.

While VT may lead to [illegible]ds to sudden cardiac death if not corrected immediately. The ventricles fibrillate, or quiver, resulting in hemodynamic collapse. Absent tissue perfusion causes end-organ ischemia. The brain is the most sensitive to this, meaning critical brain ischemia occurs quickly, and patients become unconscious, pulseless, and without respirations after a few seconds.

What is the most common clinical presentation of ventricular fibrillation (VF)?

Pathophysiology

Any pathologic process in the myocardium can produce VT and VF. Common causes include heart failure and damage from MI. These processes interrupt the normal myocyte architecture of the heart often leading to **scar tissue**.

In both VT and VF, the abnormal electrical impulse commonly originates on the border between normal and abnormal myocardium (eg, scar tissue). In VT, the electrical impulse[illegible]f the myocardium in one ventricle to the other ventricle, bypassing the His-Purkinje system, so that the QRS complex is wide (>120 msec). Put another way, the ventricles are activated in *series* rather than in *parallel*. The resulting dyssynchronous contraction, along with with the impaired diastolic ventricular filling (resulting from the fast heart rate), combine to reduce cardiac output. (It may be helpful here to review, from your basic physiology material, how a normally functioning His Purkinje system produces a narrow QRS and an efficient, synchronized muscular contraction). VF, on the other hand, which may follow VT, is merely a quivering of the ventricular chambers with no organized contraction and an immediate state of zero cardiac output.

The concept of substrate and trigger.

The pathologic processes we have described above, such as damage from MI or other forms of myocardial damage seen in heart failure, are sometimes referred to as *substrates*. The substrate is the "fertile ground" from which an arrhythmia may arise. It forms an underlying propensity for the arrhythmia. But for the arrhythmia to occur at any given time there must also be a *trigger* (something that sets it off). The trigger could be anything from an electrolyte disturbance, acute ischemia, hypoxia, stress, or a variation in the output of the autonomic nervous system. The concept of substrate and trigger can be applied to arrhythmias in general.

Monomorphic vs polymorphic VT.

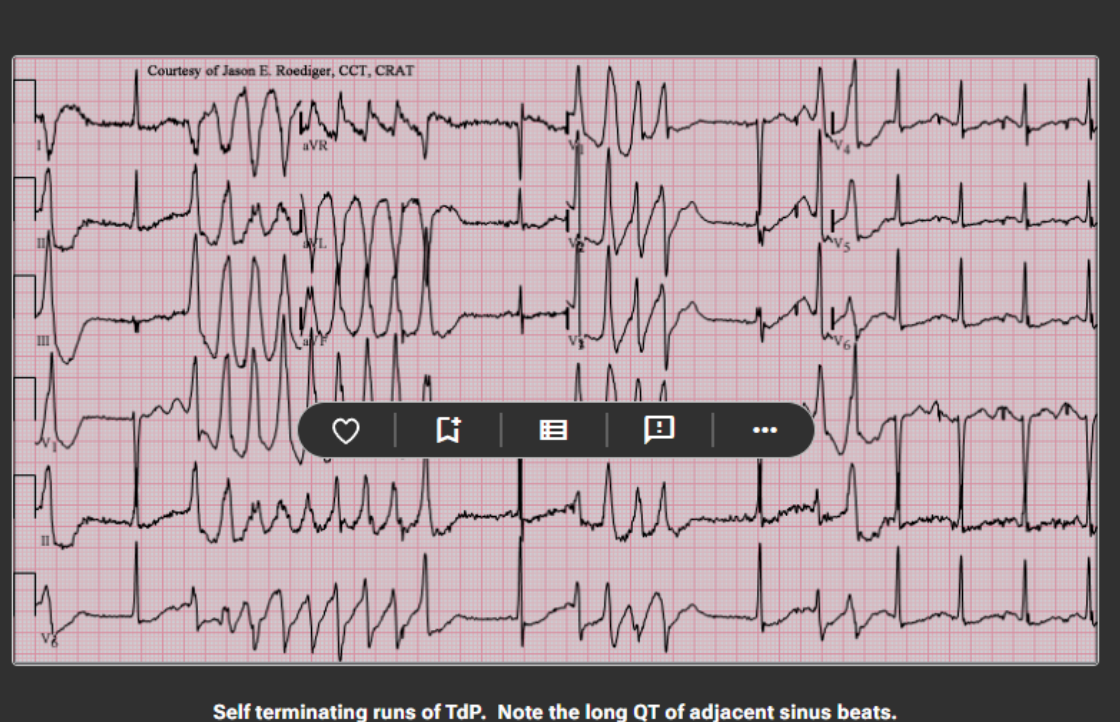
In most instances of VT all the QRS complexes are alike, which is known as *monomorphic* VT. In a minority of instances, the appearance of the QRS complexes varies, known as *polymorphic* VT. The best known and most important form of polymorphic VT is Torsades de Pointes, described below.

Torsades de Pointe

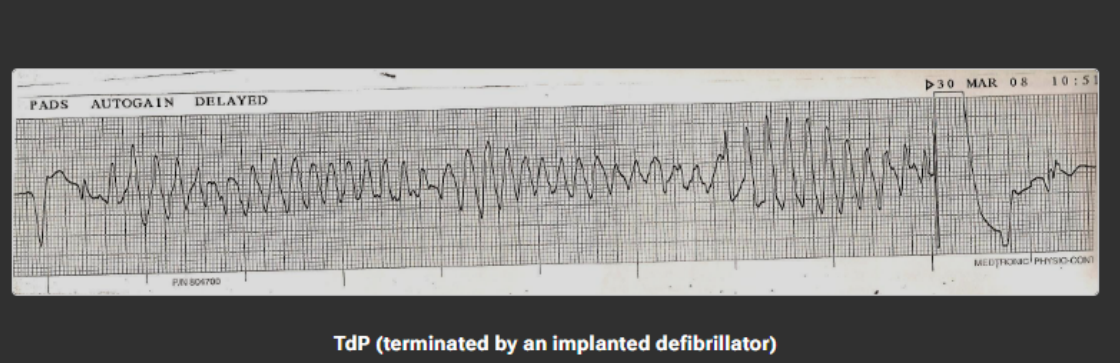
Torsades de pointes is a unique form of polymorphic VT. What distinguishes it from other forms of polymorphic VT is that it is a disorder of prolonged repolarization: prolongation of phase 3 of the action potential resulting in a long QT interval. There are two categories of long QT interval: hereditary and acquired. The hereditary form has multiple identified genes, almost all autosomal dominant. The more common acquired form can be caused by anything that prolongs action potential duration. This includes patients with potassium or magnesium depletion and ingestion of several drugs that directly alter the action potential. Torsades de pointes is characterized by irregular waveforms of alternating amplitudes and axis shifts over a range of 180 degrees, resembling twisting around an isoelectric point. TdP may present as recurring and self terminating salvos of abnormal beats.

Figure 2 illustrates the typical features: the long QT interval (must be measured in non tachycardia beats!), the shifting axis and morphology, and recurring brief salvos.

Figure 3 shows another example of TdP that required termination by an implanted defibrillator.



Self terminating runs of TdP. Note the long QT of adjacent sinus beats.



TdP (terminated by an implanted defibrillator)

Most episodes of TdP are self terminating. However, as illustrated in the figure above, some require defibrillation.

Torsades de pointes is treated with intravenous magnesium and/or potassium. When it leads to hemodynamic compromise or cardiac arrest defibrillation is necessary. The hereditary forms require long term treatment. This might consist of beta blockers or an implanted defibrillator.

CLINICAL CORRELATION

Drugs that can caus[illegible]e class IA and III antiarrhythmic drugs, macrolide and fluoroquinolone antibiotics, some antipsychotics, tricyclic antidepressants, ondansetron, methadone, and some antifungals (azoles).

Mnemonic

Long QT interval caused by drugs: ABCDEF

Anti**A**rrhythmics
Anti**B**iotics
Antipsy**C**hotics
Anti**D**epressants
Anti**E**metics
Anti**F**ungals

Brugada Syndrome.

The second most common inherited disorder of the action potential that can cause lethal ventricular arrhythmias is Brugada syndrome, a disorder of sodium channels. Most patients are typically asymptomatic until they present in sudden cardiac arrest from VT or VF. Therefore early diagnosis, based on recognition of characteristic ECG findings, is essential. Inheritance is primarily autosomal dominant with variable expression. For some cases, a gene has not yet been identified. [illegible]l spontaneous but can be triggered by fever, cocaine, or tricyclic antidepressant use. Detailed discussion of Brugada syndrome is beyond the discussion of this brick.

Diagnosis of VT and VF on the electrocardiogram

VT and VF are diagnosed on ECG. VT shows a series of wide QRS complexes at 100 to 250 beats/min, three or more in a row. (You can think of it as multiple PVCs in a row). As stated above, VT can be monomorphic (QRS complexes of the same shape) or polymorphic (QRS complexes of different shapes). P waves, if seen at all, will bear no relationship to the QRS complexes.

Figure 4 below shows a monomorphic pattern.

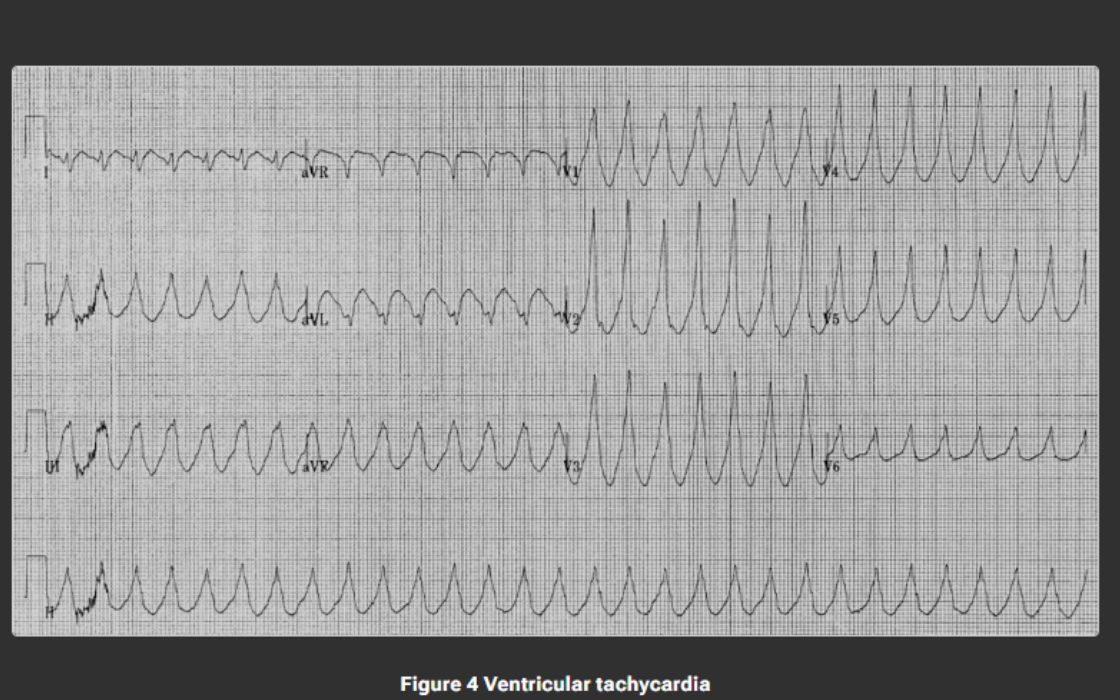


Figure 4 Ventricular tachycardia

In contrast, VF has no organized QRS complexes. The appearance is that of random oscillations in the tracing, sometimes in the form of a fine "quivering", in much the same way the heart muscle is quivering. See Figure 5 below.

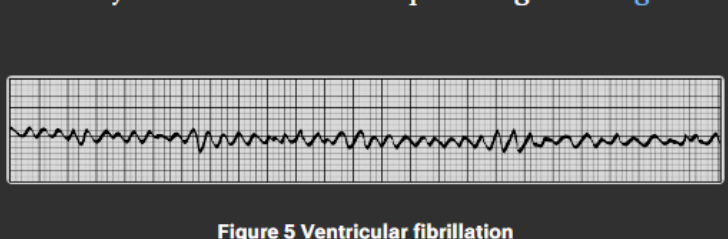


Figure 5 Ventricular fibrillation

Management of Ventricular Tachycardia

Although patients might remain conscious with VT, it should be treated as a medical emergency because it is unstable and rapidly leads to compromised blood flow. If the patient has a pulse but is hypotensive, **synchronized cardioversion** is used. This allows the defibrillator to read the current electrical pattern of the heart and apply a shock at a particular moment to increase the probability that the patient's heart will return to sinus rhythm.

If the patient is pulseless, the treatment is immediate **unsynchronized defibrillation** and the basic cardiac life support (BCLS) protocol (ie, chest compressions). Unsynchronized defibrillation is a high-voltage energy shock that is not coordinated with the heartbeat.

If the patient is stable, medications can be used to terminate VT or prevent recurrence. These include Amiodarone and Lidocaine.

Management of Ventricular Fibrillation

VF is always unstable and emergent. It is treated similarly to pulseless VT, with defibrillation.

CLINICAL CORRELATION

The training of the lay public in the use of defibrillators (now present in many public areas) is an effective public health measure to save the lives of patients with VT or VF. In addition, the use of implantable cardioverter-defibrillators that sense and treat these arrhythmias is effective in many patients with systolic heart failure and after MI.

Prevention

Many patients with VT/VF should receive an **implantable cardioverter-defibrillator** (**ICD**), which contains pacing/sensing electrodes, a defibrillator, and a pulse generator. In most cases an implanted defibrillator is combined with a pacemaker.

Figure 6 below is a cartoon representation of a dual chamber (right atrial and right ventricular) pacemaker. Defibrillator capability is contained in specialized portions of the leads (shock coils) not shown in the figure. Figure 7 below it is a chest xray image of a single chamber (right ventricular) pacemaker combined with defibrillation capability. The defibrillator electrodes are specialized portions (shock coils) of the pacing wire seen as the more dense thickened portions of the lead.

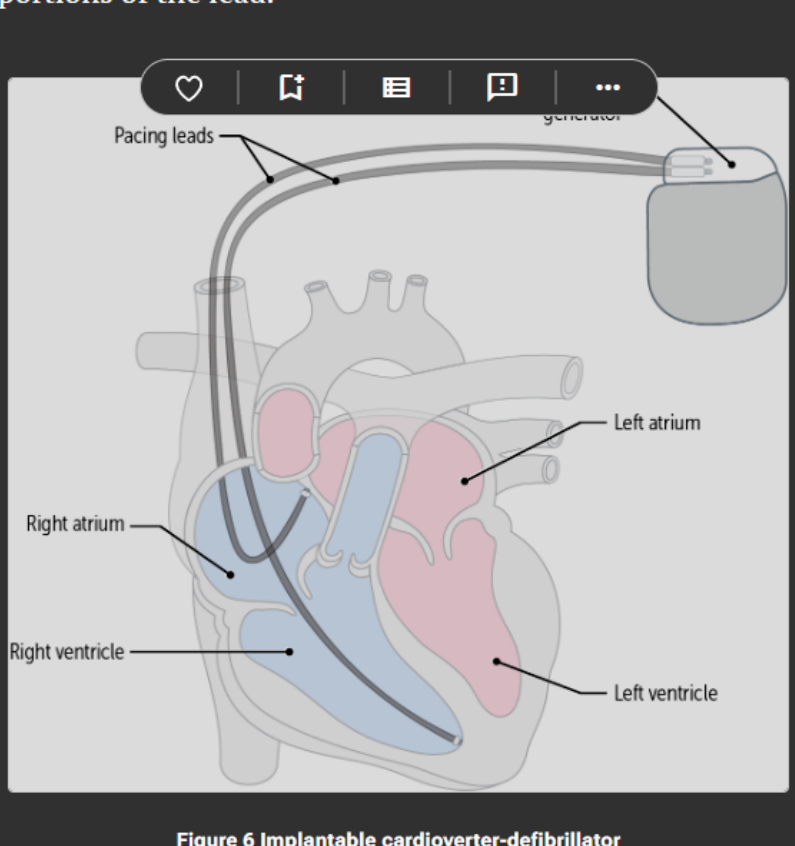


Figure 6 Implantable cardioverter-defibrillator

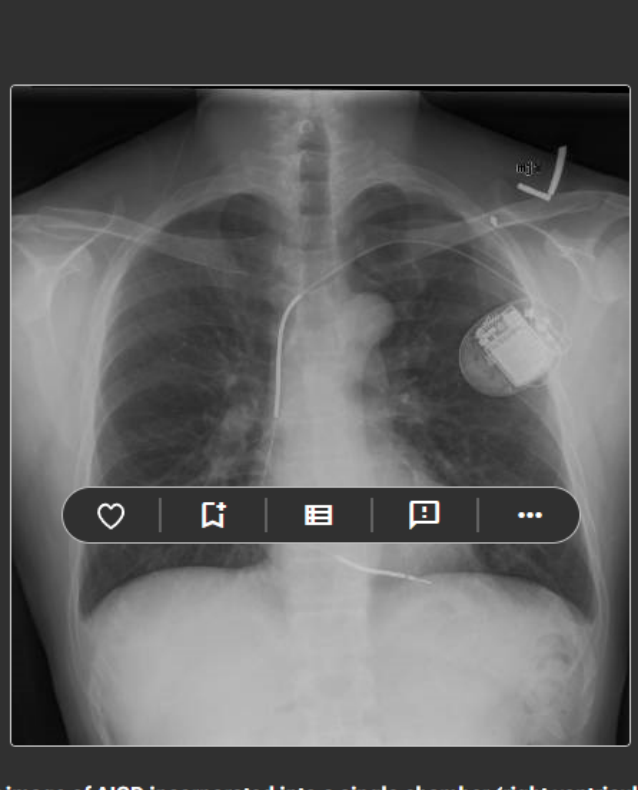


Figure 7. Xray image of AICD incorporated into a single chamber (right ventricular) pacemaker

Indications for ICD placement include VF/VT in the setting of most types of structural heart disease like cardiomyopathy, valve disease, or left ventricular hypertrophy (LVH). Exceptions are those who only have VT/VF in the first 48 hours after a MI, or in whom a reversible noncardiac cause like a drug overdose or electrolyte imbalance was the cause of the VF/VT. Many such devices are implanted after MI in patients with low EF who have never had an arrhythmia, just as a prevention of sudden cardiac death (aka *primary prevention* of SCD).

What Is Accelerated Idioventricular Rhythm?

Some ventricular tachycardias are slower than others. When the rate of repetitive ventricular depolarizations gets as low as 100 beats per minute the arrhythmia is referred to as accelerated ventricular rhythm (AIVR) rather than ventricular tachycardia. The range of rates that define this particular arrhythmia is 60 to 100 beats per minute. If the rate is below 60 it is referred to as merely idioventricular rhythm (IVR).

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to GJ, what is the next step in his treatment? How do you explain to him what happened?

GJ is pulseless and receives basic life support with defibrillation. He is in ventricular tachycardia (VT) and undergoes successful cardioversion of normal sinus rhythm and is transferred to the coronary care unit. You visit GJ and explain to him that he had an episode of VT, a serious abnormal heart rhythm that begins in the lower chambers of the heart. Because the heart rate is so high, the heart can't pump efficiently and blood backs up into the lungs, causing shortness of breath. You tell GJ that VT can occur after a CABG, and he'll be staying in the ICU for a few days for further monitoring.

Summary

- Premature ventricular contractions (PVCs) are common arrhythmias that appear as extra heartbeats and are typically benign.
- Ventricular tachycardia (VT) and ventricular fibrillation (VF) are dangerous arrhythmias often resulting from myocardial infarction, characterized by wide QRS complexes and lack of organized QRS complexes, respectively.
- Torsades de pointes is a form of VT associated with long QT syndrome (hereditary form) or drugs and electrolyte imbalances (acquired forms).
- Brugada syndrome is a genetic disorder that predisposes individuals to VT/VF, often triggered by fever or drug overdose.
- Accelerated idioventricular rhythm (AIVR) occurs during myocardial ischemia or infarction, with wide QRS complexes and a rate of 60 to 100 beats/min, typica[illegible]nic treatment.

Review Questions

1. An 80-year-old male is admitted to the intensive care unit with chest pain. His initial ECG showed ST-segment elevations in leads II, III, and aVF. He becomes unresponsive, and the nurse cannot feel a pulse. Which of the following arrhythmias is most likely present?

A. ○ Accelerated idioventricular rhythm
B. ○ Idioventricular rhythm
C. ○ Junctional rhythm
D. ○ Supraventricular tachycardia
E. ● Ventricular tachycardia

Hide Explanation

The correct answer is ventricular tachycardia (E). In this arrhythmia, the ventricles contract at such a rapid rate that they cannot fill properly, leading to a drop in cardiac output, and sometimes pulseless arrest. Accelerated idioventricular rhythm (A) and idioventricular rhythm (B) are usually asymptomatic, although slow idioventricular rhythm with pulses of 30 to 40 beats/min may lead to dizziness or syncope. Junctional rhythm (C) is usually asymptomatic. Supraventricular tachycardia (D) can present as a pulseless cardiac arrest but more commonly leads to dizziness, heart fa[illegible]

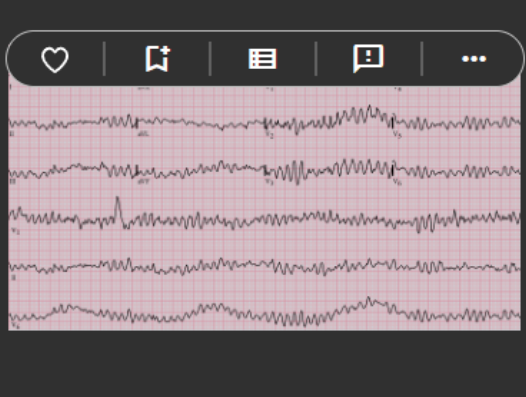
2. What is a possible sign or symptom of ventricular tachycardia?

A. ○ Chest pain
B. ○ Syncope
C. ○ Dyspnea
D. ○ Palpitations
E. ● All of the above

Hide Explanation

The correct answer is all of the above (E). VT can produce any type of cardiovascular symptom. The tachycardic rate increases oxygen consumption which can lead to ischemia and consequently chest pain. Cardiac output falls due to both dyssynchronous ventricular contraction and reduced diastolic filling time. This can lead to dizziness or syncope. Impaired diastolic filling and myocardial ischemia may produce increases in pulmonary capillary wedge pressure leading to dyspnea. Patients who become aware of the rapid heart rate may experience palpitations.

3. What is the treatment for the condition shown below?



A. ○ Observation and monitoring
B. ● Unsynchronized defibrillation
C. ○ Synchronized cardioversion
D. ○ Intravenous diltiazem

Hide Explanation

The condition shown is ventricular fibrillation. The correct answer is B, unsynchronized defibrillation. Answer (A) is incorrect because death is certain without prompt treatment. Answer (C), synchronized cardioversion is incorrect, as synchronization of the device with the rhythm is impossible in the absence of a QRS complex. Given the need for immediate restoration of normal rhythm, electricity, not pharmacology, is necessary, making answer (D) incorrect. Moreover, calcium blockers are generally contraindicated in ventricular arrhythmias.

4. Which agent can be used to terminate a ventricular tachycardia if the patient is hemody[illegible]

A. ○ Adenosine
B. ○ Atropine
C. ● Amiodarone
D. ○ Atenolol

Hide Explanation

The correct answer is (C) Amiodarone. Amiodarone is an antiarrhythmic drug (Class III) that blocks potassium channels and prolongs the refractory period. It is commonly used for the termination of ventricular tachycardia (VT) in hemodynamically stable patients. It can be given intravenously for acute management. Incorrect response (A) Adenosine: Primarily used for supraventricular tachycardia (SVT) due to its ability to transiently block AV nodal conduction. It is not effective for VT, which originates in the ventricles and does not depend on AV nodal conduction. Incorrect response (B) Atropine: A muscarinic antagonist that increases heart rate by blocking vagal tone. It is indicated for bradyarrhythmias, not tachyarrhythmias. Using atropine in VT would worsen the condition. Incorrect response (D) Atenolol: A beta-blocker that slows AV nodal conduction and reduces heart rate. While beta-blockers can help prevent arrhythmias or control rate in certain conditions, they are not first-line for terminating VT acutely.

How would you rate this Brick?

References

- University of Utah ECG Learning Center, Section on VT https://ecg.utah.edu/lesson/5-3#vent_tachy
- Statpearls review on VT https://www.ncbi.nlm.nih.gov/books/NBK532954/

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Describe the pathophysiology of ventricular arrhythmias~~
02 ☑ ~~Describe the clinical findings associated with ventricular arrhythmias~~
03 ☑ ~~Recognize the characteristic features of QTc prolongation~~
04 ☑ ~~Outline a treatment plan for ventricular arrhythmias~~

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Objective Confidence: 4/4